Computer Number Systems

Different Ways To Say How Many

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Learning Outcomes

- Explain why computer designers chose to use the binary system for representing information in computers.
- Explain what a “binary digit” is.
- Explain what a “byte” is.

Learning Outcomes

- Computer Number Systems
  - Convert decimal numbers to binary.
  - Convert binary numbers to decimal.
  - Convert binary numbers to hexadecimal.
  - Convert hexadecimal. Numbers to binary.
  - Convert hexadecimal numbers to decimal.
  - Convert decimal numbers to hexadecimal.

Learning Outcomes

- Associate electronic prefixes with their meanings.
- Identify the special quantities specified by the terms “kilobyte” and “megabyte.”

Learning Outcomes

- Identify the special code used to represent alphanumeric characters in PCs.
- Describe the parity method of detecting data errors in PCs.

Why binary?

- The original computers were designed to be high-speed calculators.
- The designers needed to use the electronic components available at the time.
- The designers realized they could use a simple coding system--the binary system--to represent their numbers

Representing Information in Computers

- All the different types of information in computers can be represented using binary code.
  - Numbers
  - Letters of the alphabet and punctuation marks
  - Microprocessor instruction
  - Graphics/Video
  - Sound

Bits and Bytes

- A binary digit is a single numeral in a binary number.
- Each 1 and 0 in the number below is a binary digit:
  - 1 0 0 1 0 1 0 1
- The term "binary digit" is commonly called a "bit."
- Eight bits grouped together is called a "byte."

Computer Number Systems

- Decimal Numbers
- Binary Numbers
- Hexadecimal Numbers

Decimal Number System

- The prefix “deci-” stands for 10
- The decimal number system is a Base 10 number system:
  - There are 10 symbols that represent quantities:
    - 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
  - Each place value in a decimal number is a power of 10.

Background Information

- Any number to the 0 (zero) power is 1.
  - $4^0 = 1$ $16^0 = 1$ $1{,}482^0 = 1$.
- Any number to the 1st power is the number itself.
  - $10^1 = 10$ $49^1 = 49$ $827^1 = 827$

Numbering Systems

Decimal Number System

10^3	10^2	10^1	10^0
1000	100	10	1
1	4	9	2

Numbering Systems

Decimal Number System

1492

1 x	1000 =	1000
4 x	100 =	400
9 x	10 =	90
2 x	1 =	+ 2
		1492

Binary Numbers

- The prefix “bi-” stands for 2
- The binary number system is a Base 2 number system:
  - There are 2 symbols that represent quantities:
    - 0, 1
  - Each place value in a binary number is a power of 2.

Numbering Systems

Binary Number System

8	4	2	1
2^3	2^2	2^1	2^0
1	0	1	1

Numbering Systems

Binary Number System

1011

1 x 8 = 8

0 x 4 = 0

1 x 2 = 2

1 x 1 = + 1

11

Numbering Systems

Binary Number System

128	64	32	16	8	4	2	1
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1	0	1	1	0	1	0	1

Converting Binary Numbers to Decimal

- Step 1
  - Starting with the 1s place, write the binary place value over each digit in the binary number being converted.

16	8	4	2	1
1	0	1	0	1

Converting Binary Numbers to Decimal

- Step 2
  - Add up all of the place values that have a “1” in them.

16 8 4 2 1

1 0 1 0 1

16 + 4 + 1 = 21

You Try It!

- Convert the binary number 1 1 0 0 1 0 1 to decimal.

Step 1

64	32	16	8	4	2	1
1	1	0	0	1	0	1

Step 2 64 + 32 + 4 + 1=101

Converting Decimal Numbers to Binary

- There are two methods that can be used to convert decimal numbers to binary:
  - Repeated subtraction method
  - Repeated division method
- Both methods produce the same result and you should use whichever one you are most comfortable with.

Converting Decimal Numbers to Binary

- The Repeated Subtraction method

- As an explanation of the repeated subtraction method, let’ s convert the decimal number **853** to binary.

Converting Decimal Numbers to Binary

- The Repeated Subtraction method
  - Step 1:
    - Starting with the 1s place, write down all of the binary place values in order until you get to the first binary place value that is GREATER THAN the decimal number you are trying to convert.



1024 512 256 128 64 32 16 8 4 2 1

Converting Decimal Numbers to Binary

- The Repeated Subtraction method
  - Step 2:
    - Mark out the largest place value (it just tells us how many place values we need).

853

~~1024~~ 512 256 128 64 32 16 8 4 2 1

Converting Decimal Numbers to Binary

- The Repeated Subtraction method
  - Step 3:
    - Subtract the largest place value from the decimal number. Place a “1” under that place value.

853 - 512 = 341

512	256	128	64	32	16	8	4	2	1
1									

Converting Decimal Numbers to Binary

- Step 4:
  For the rest of the place values, try to subtract each one from the previous result.
  - If you can, place a “1” under that place value.
  - If you can’t, place a “0” under that place value.

Converting Decimal Numbers to Binary

- The Repeated Subtraction method
  - Step 5:
    - Repeat Step 4 until all of the place values have been processed.
  - The resulting set of 1s and 0s is the binary equivalent of the decimal number you started with.

Converting 853 to Binary



You Try It!

- Convert the decimal number 587 to binary.

- 1 0 0 1 0 0 1 0 1 1

Converting Decimal Numbers to Binary

- The Repeated Division method
- The general technique of this method can be used to convert any decimal number to any other number system.

Converting Decimal Numbers to Binary

- Step 1:
  - Divide the decimal number you're trying to convert by 2 in regular long division until you have a final remainder.
- Step 2:
  - Use the remainder as the LEAST SIGNIFICANT DIGIT of the binary number.

Converting Decimal Numbers to Binary

- Step 3:
  - Divide the quotient you got from the first division operation until you have a final remainder.
- Step 4:
  - Use the remainder as the next digit of the binary number.

Converting Decimal Numbers to Binary

- Step 5:
  - Repeat Steps 3 & 4 as many times as necessary until you get a quotient that can't be divided by 2.
- Step 6:
  - Use the last remainder (the one that can't be divided by 2) as the MOST SIGNIFICANT digit.

An Example of the Repeated Division Method

- This example converts 853 to binary (the same example we used for the repeated subtraction method).
- Step 1:
  - 853 / 2 = 426 Remainder 1
- Step 2:
  - The remainder of 1 becomes the LEAST significant digit of the number.
  -

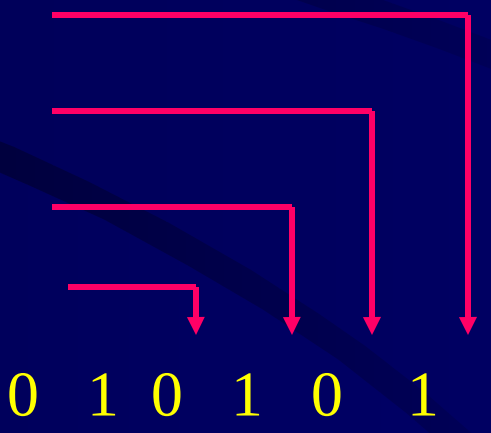


An Example of the Repeated Division Method

- Step 3:
  - Divide the quotient from Step 1 by 2 all the way out.
  - 426 / 2 = 213 Remainder 0
- Step 4:
  - The remainder of 0 becomes the next digit of the number.
  - 0 1

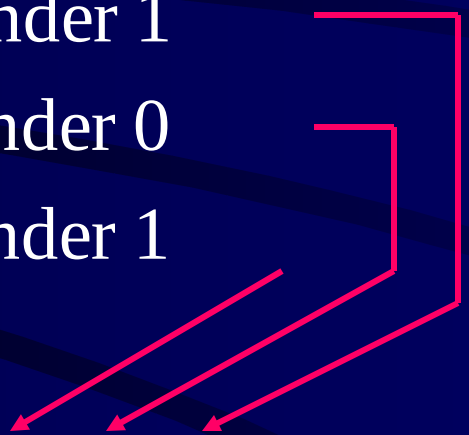
An Example of the Repeated Division Method

- Step 5:
  - Continue to divide the quotients by 2 and move the remainders down until you get a quotient that can't be divided by 2.
    - 213 / 2 = 106 Remainder 1
    - 106 / 2 = 53 Remainder 0
    - 53 / 2 = 26 Remainder 1
    - 26 / 2 = 13 Remainder 0
    -



An Example of the Repeated Division Method

- Step 5 (Continued):
  - 13 / 2 = 6 Remainder 1
  - 6 / 2 = 3 Remainder 0
  - 3 / 2 = 1 Remainder 1
  - 1 1 0 1 0 1 0 1 0 1
- Step 6:
  - The final quotient of 1 comes down to be the most significant digit.



Hexadecimal Numbers

- The prefix “hexa-” stands for 6 and the prefix “deci-’ stands for 10
- The hexadecimal number system is a Base 16 number system:
  - There are 16 symbols that represent quantities:
    - 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
  - Each place value in a hexadecimal number is a power of 16.

Hexadecimal Numbers

- We use hexadecimal numbers as “shorthand” for binary numbers
- Each group of four binary digits can be represented by a single hexadecimal digit.

Hexadecimal Numbers

Dec	Bin	Hex	Dec	Bin	Hex
0	0000	0	8	1000	8
1	0001	1	9	1001	9
2	0010	2	10	1010	A
3	0011	3	11	1011	B
4	0100	4	12	1100	C
5	0101	5	13	1101	D
6	0110	6	14	1110	E
7	0111	7	15	1111	F

Converting Binary Numbers to Hexadecimal

- Step 1:
  - Starting with the LEAST SIGNIFICANT digit, mark off the digits in groups of 4.
  - For example, to convert 110001011011 to hexadecimal, mark off the digits in groups of four.
    - 1 1 0 0 | 0 1 0 1 | 1 0 1 1

Converting Binary Numbers to Hexadecimal

- Step 2:
  - Convert each group of four digits to its hexadecimal character.
    - 1 1 0 0 | 0 1 0 1 | 1 0 1 1
    - C 5 B

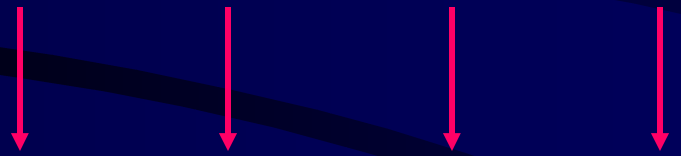
Converting Binary Numbers to Hexadecimal

- Helpful Hint
  - The last group on the left can have anywhere from 1 to 4 binary digits group.
  - If it will help you see the pattern, you can fill in enough leading zeroes to make the last group on the left have four digits.
  - For example, 1 1 0 | 0 1 1 1 | 1 0 0 1 could be written 0 1 1 0 | 0 1 1 1 | 1 0 0 1

Converting Hexadecimal Numbers to Binary

- Converting hexadecimal numbers to binary is just the reverse operation of converting binary to hexadecimal.
- Just convert each hexadecimal digit to its four-bit binary pattern. The resulting set of 1s and 0s is the binary equivalent of the hexadecimal number.

Example of Hexadecimal to Binary Conversion

- Convert A3D7 to binary.
- A 3 D 7
- 
- 1010 0011 1101 0111

Decimal to Hexadecimal Conversion

- There are two methods to choose from:
  - Do a decimal-to-binary conversion and then a binary-to-hexadecimal conversion.
  - Do a direct conversion using the repeated division method.
    - Since this is a conversion to hexadecimal, 16 is the divisor each time.

- This example converts 853 to hexadecimal.
- Step 1:
  - 853 / 16 = 53 Remainder 5
- Step 2:
  - The remainder of 5 becomes the LEAST significant digit of the number.
  - 5



An Example of the Repeated Division Method

- Step 3:
  - Divide the quotient from Step 1 by 2 all the way out.
  - 53 / 16 = 3 Remainder 5
- Step 4:
  - The remainder of 5 becomes the next digit of the number.
  - 5 5

An Example of the Repeated Division Method

- Step 5:
  - The final quotient of 3 comes down to be the most significant digit.

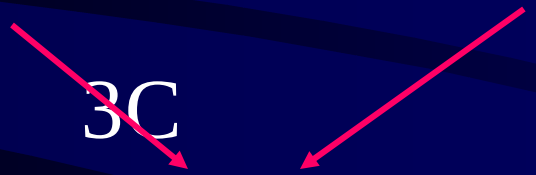


  - 3 5 5
- So, the hexadecimal equivalent of 853 is 355.

Decimal to Hexadecimal Conversion

- Note:
  - Since you are dividing by 16 in the repeated division method for decimal-to-hex conversion, you could end up with remainders of anywhere from 0 to 15.
  - If a remainder is 10 to 15, you convert it to the single hex symbol when you add the digit to the hex number you're building.

Another Decimal-to-Hex Example

- Let’s convert decimal 60 to hexadecimal.
- 60 /16 = 3 Remainder 12
- 

  - The remainder of 12 is represented by its hex symbol “C” in the resulting number and the quotient of “3” can’t be divided by 16 so it comes down to be the most significant digit of the hex number.

Hexadecimal Number System

16^3 16^2 16^1 16^0

4096 256 16 1

2 F A 4

Converting Hexadecimal Numbers to Decimal

- Multiply each digit of the hex number by its place value and add the results.
- For example, converting 2FA4
  - 2 x 4096 = 8192
  - 15 x 256 = 3840 (convert "F" to 15)
  - 10 x 16 = 160 (convert "A" to 10)
  - 4 x 1 =+ 4
  - 12,196

Electronics Prefixes

- There is a set of of terms used in electronics used to represent different powers of ten.
- There is a set of terms used to represent large whole numbers and a set of terms used to represent small fractional numbers.

Electronics Prefixes for Large Whole Numbers

Prefix	Value	Abbreviation
Kilo	1,000	K
Mega	1,000,000	M
Giga	1,000,000,000	G
Tera	1,000,000,000,000	T

Electronics Prefixes for Small Fractional Numbers

Prefix	Value	Abbreviation
milli	1 / 1,000	m
micro	1 / 1,000,000	μ
nano	1 / 1,000,000,000	n
pico	1 / 1,000,000,000,000	p

Exceptions to the Rule

- A “kilobyte” (KB) is 1,024 bytes.
- A “megabyte” (MB) is 1,048,576 bytes.
- These values come from the nearest binary place values to 1,000 and 1,000,000.

A Code for Letters and Symbols

- PCs use a standard binary code to represents letters of the alphabet, numerals, punctuation marks and other special characters.
- The code is called ASCII (pronounced “as-key”) which stands for “American Standard Code for Information Interchange.”
- There are 256 code combinations.

Examples of ASCII Representation

Character	Code	Hex
A	0100 0001	41
?	0011 1111	3F
8	0011 1000	38
space	0010 0000	20

A Method for Detecting Errors

- When all the information is represented by binary numbers, accuracy of each binary digit is absolutely essential.
- A change of just one bit in a byte can completely change the meaning of the byte.

Examples of Binary Errors

Original Code	Meaning	Changed Code	Meaning
0100 0111	“G”	0100 0011	“C”
1000 0011	131	0000 0011	3
1001 1110	Add	1001 1111	Subtract

The Parity Method for Detecting Errors

- A special circuit counts the number of 1 bits in a byte and adds a special ninth bit called the “parity bit.”
- When the stored byte is later read out, the parity checking circuit re-counts the number of 1 bits and check for the correct number.

Parity Bits

- There are two methods for checking parity:
  - Odd
  - Even
- Both methods are equally effective but the method must be consistent within an operation.
  - If odd parity was used to store the byte, odd parity must be used to read it.

Odd Parity

- The parity checking circuit counts the number of 1 bits and adds the parity bit to make the total number of 1 bits an ODD number.
- Examples
  - 1 0 0 1 0 0 1 1 has four 1s so the parity bit would be a 1
  - 1 1 0 0 1 1 1 0 has five 1s so the parity bit would be a 0

Even Parity

- The parity checking circuit counts the number of 1 bits and adds the parity bit to make the total number of 1 bits an EVEN number.
- Examples
  - 1 0 0 1 0 0 1 1 has four 1s so the parity bit would be a 0
  - 1 1 0 0 1 1 1 0 has five 1s so the parity bit would be a 1

Summary